LESSON 01 / 17

Accumulation begins with a partition

For a bounded function f on $[a, b]$, divide the interval into n equal pieces of width

$$\Delta x = \frac{b-a}{n}, \qquad x_i = a + i\Delta x.$$

The right-endpoint Riemann sum is

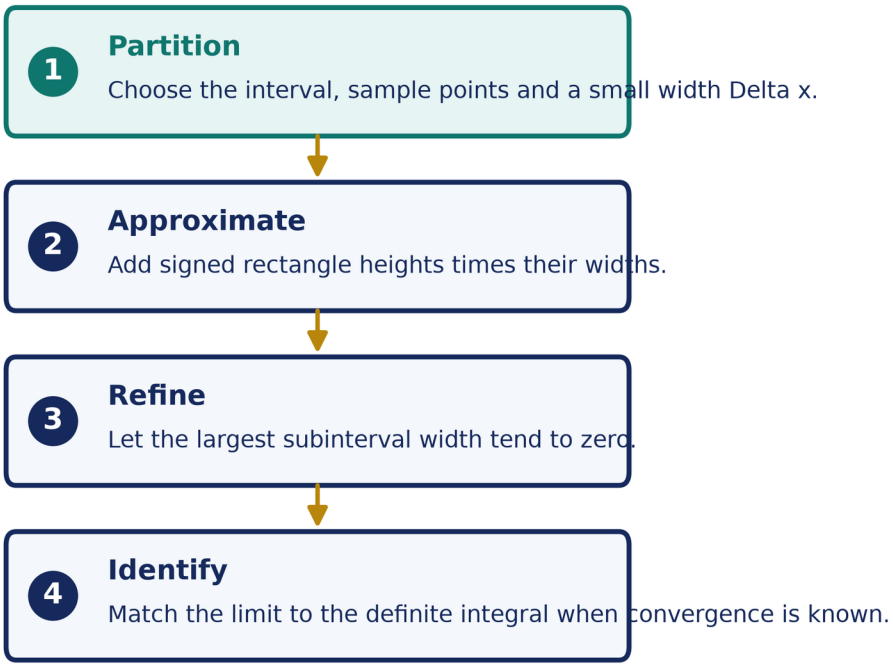
$$R_n = \sum_{i=1}^{n} f(x_i)\Delta x.$$

For continuous f, these sums approach the definite integral:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right).$$

The integral is a **signed** accumulator: heights below the x-axis contribute negatively.

A Riemann sum becomes an integral only in the limit



READ THE FIGURE

A finite rectangle sum is an approximation or a discrete identity. The limiting process is what defines the continuous accumulation.

WORKED EXAMPLE

Evaluate the source's x-squared area directly as a limit

For $f(x)=x^2$ on $[0,2]$, $\Delta x = 2/n$ and $x_i = 2i/n$. Therefore

$$R_n = \frac{2}{n}\sum_{i=1}^{n}\left(\frac{2i}{n}\right)^2 = \frac{8}{n^3}\sum_{i=1}^{n} i^2.$$

Using $\sum_{i=1}^{n} i^2 = n(n+1)(2n+1)/6$,

$$R_n = \frac{4(n+1)(2n+1)}{3n^2} \longrightarrow \frac{8}{3}.$$

Thus $\int_0^2 x^2\,dx = 8/3$. The editable visual lab recreates the source's right-endpoint rectangles.

A Riemann sum can also be the unknown in a limit

If a limit has the form

$$\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n} f\left(\frac{i}{n}\right),$$

then it equals $\int_0^1 f(x)\,dx$, provided f is integrable. For example,

$$\lim_{n\to\infty}\frac{1}{n}\sum_{i=1}^{n}\frac{i}{n} = \int_0^1 x\,dx = \frac{1}{2}.$$

COMMON TRAP

Do not call signed area by the shorter word area alone

On $[-1,1]$, $\int_{-1}^{1} x^3\,dx = 0$, but the geometric area between the cubic and the axis is $1/2$. Total area requires $\int_a^b |f(x)|\,dx$ and a split at zeros.

LESSON 02 / 17

Accumulated area differentiates back to the height

For continuous f, define

$$A(x) = \int_a^x f(t)\,dt.$$

Then

$$A(x+h) - A(x) = \int_x^{x+h} f(t)\,dt.$$

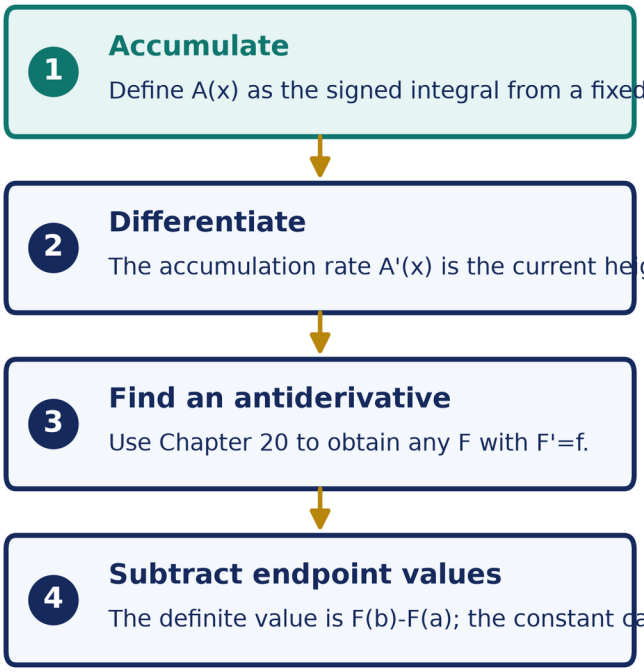
For small h, the continuous height is close to $f(x)$, so the difference quotient tends to $f(x)$:

$$A'(x) = f(x).$$

This is FTC Part 1. If $F' = f$, FTC Part 2 gives

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

The Fundamental Theorem joins the two directions of calculus



READ THE FIGURE

The dummy variable t prevents confusion with the moving upper limit x. An antiderivative's arbitrary constant does not affect a definite difference.

WORKED EXAMPLE

S9 and Q23: subtract antiderivative values

$$\int_0^1 x^2\,dx = \left[\tfrac{x^3}{3}\right]_0^1 = \tfrac{1}{3}, \qquad \int_0^1 x^3\,dx = \left[\tfrac{x^4}{4}\right]_0^1 = \tfrac{1}{4}.$$

The first result agrees with a right-sum limit on $[0,1]$; the second also equals the total area because $x^3 \geq 0$ on this interval.

WORKED EXAMPLE

S10 and Q22: differentiate an accumulation function

If $A(x) = \int_0^x t^2\,dt$, then $A'(x) = x^2$. Indeed $A(x) = x^3/3$. Similarly,

$$\frac{d}{dx}\int_0^x t^3\,dt = x^3,$$

because the explicit accumulation function is $x^4/4$.

Variable upper and lower limits use the chain rule

For differentiable u, v and continuous f,

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt = f(v(x))v'(x) - f(u(x))u'(x).$$

For example,

$$\frac{d}{dx}\int_{x^2}^{\sin x} e^{t^2}\,dt = e^{\sin^2 x}\cos x - 2xe^{x^4}.$$

No elementary antiderivative of e^{t^2} is needed.

COMMON TRAP

Do not differentiate the integrand with respect to x

In $\int_0^x f(t)\,dt$, the integrand uses the dummy variable t. FTC substitutes the upper limit into f; it does not produce $\int_0^x f'(t)\,dt$ or $f'(x)$.

LESSON 03 / 17

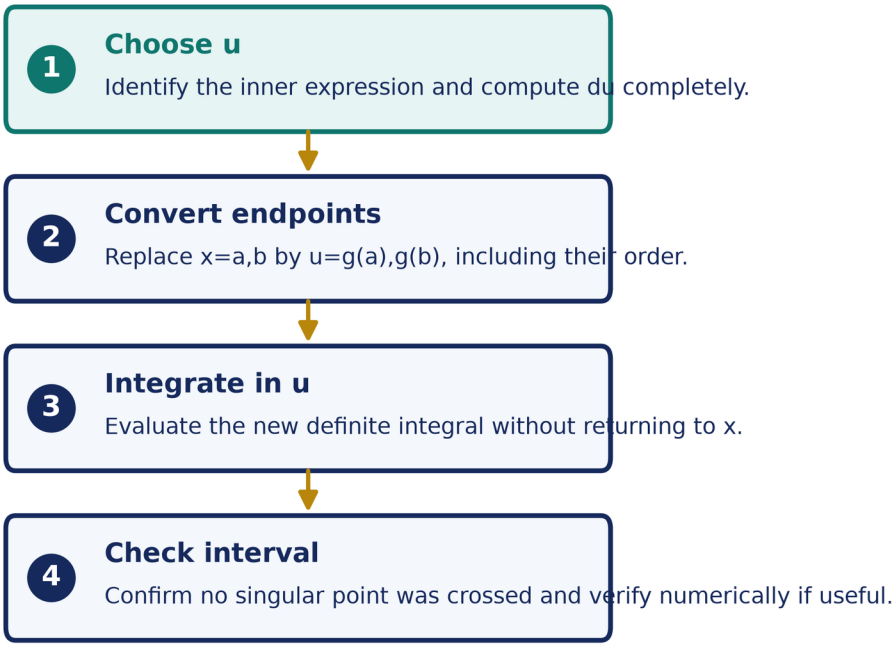
Either transform the limits or return to x before evaluating

When $u = g(x)$ is used in a definite integral, the endpoints become $g(a)$ and $g(b)$:

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

The substitution must be valid on the integration interval; monotonicity is a convenient sufficient condition but is not the only possible justification.

A definite substitution changes the whole problem



READ THE FIGURE

Alternatively, find an x-antiderivative and use the original endpoints. Do not mix u-limits with an x-antiderivative.

WORKED EXAMPLE

P4 and Q7: an exponential substitution

For $u = x^2$, $du = 2x\,dx$, and $x = 0, 1$ become $u = 0, 1$:

$$\int_0^1 2xe^{x^2}dx = \int_0^1 e^u du = e - 1.$$

The numerical value is $1.718281828\ldots$.

WORKED EXAMPLE

PS and QII: complete the square and translate both endpoints

Since $x^2+2x+2=(x+1)^2+1$, put $u=x+1$. The limits $0,1$ become $1,2$:

$$\int_0^1 \frac{dx}{x^2+2x+2}=\int_1^2 \frac{du}{u^2+1}=\tan^{-1}(2)-\frac{\pi}{4}.$$

This is approximately 0.3217505544.

Two standard-form evaluations

For Q10 and Q12,

$$\int_0^1 \frac{dx}{1+x^2}=\tan^{-1}(1)-\tan^{-1}(0)=\frac{\pi}{4},$$

$$\int_0^1 \frac{dx}{\sqrt{4-x^2}}=\left[\sin^{-1}\frac{x}{2}\right]_0^1=\frac{\pi}{6}.$$

The second integrand is continuous on $[0,1]$, so no improper endpoint issue arises.

COMMON TRAP

Reversed u-limits carry a sign

If $g(a)>g(b)$, keep that order. Reversing the new limits changes the sign. Also, an antiderivative cannot be used across an interior singularity of the original integrand.

LESSON 04 / 17

The Chapter 20 toolkit becomes an endpoint calculation

The definite method is unchanged: prepare, integrate and then subtract endpoint values. Keep any interval restriction visible until the final evaluation.

WORKED EXAMPLE

Q17: by parts on a trigonometric interval

$$\int_0^{\pi} x\sin x\,dx = [-x\cos x + \sin x]_0^{\pi} = (\pi + 0) - (0 + 0) = \pi.$$

Although $x\sin x \geq 0$ on $[0, \pi]$, this value is also the geometric area under that arch.

WORKED EXAMPLE

Q20: partial fractions with logarithmic endpoint values

Using the Chapter 20 decomposition,

$$\int_0^1 \frac{dx}{(x+1)(x+2)} = \left[\ln\left|\frac{x+1}{x+2}\right|\right]_0^1 = \ln\frac{2}{3} - \ln\frac{1}{2} = \ln\frac{4}{3}.$$

The interval $[0, 1]$ avoids both poles -2 and -1.

The half-angle substitution needs valid new limits

WORKED EXAMPLE

S13 and Q35: the complete definite calculation

For $t = \tan(x/2)$ on $[0, \pi/2]$,

$$\cos x = \frac{1-t^2}{1+t^2}, \qquad dx = \frac{2dt}{1+t^2},$$

and the limits become $0 \to 0$ and $\pi/2 \to 1$. Hence

$$\int_0^{\pi/2} \frac{dx}{2+\cos x} = \int_0^1 \frac{2}{3+t^2}dt = \frac{2}{\sqrt{3}}\left[\tan^{-1}\frac{t}{\sqrt{3}}\right]_0^1.$$

Since $\tan^{-1}(1/\sqrt{3}) = \pi/6$,

$$I = \frac{\pi}{3\sqrt{3}} \approx 0.6045997881.$$

Three definite techniques, three independent endpoint checks

Question	Antiderivative or transformed form	Value
Q17	-x cos x + sin x	pi
Q20	ln|(x+1)/(x+2)|	ln(4/3)
Q35	2 arctan(t/sqrt(3))/sqrt(3)	pi/(3 sqrt(3))

READ THE FIGURE

The technique is selected from the integrand. The definite value is then forced by endpoints or transformed limits, not by memorising a numerical answer.

COMMON TRAP

An antiderivative on a branch is enough only if the whole interval stays on that branch

The half-angle calculation uses $0 \le x \le \pi/2$, safely inside $(-\pi, \pi)$. A larger interval reaching an odd multiple of π would require branch and convergence analysis first.

LESSON 05 / 17

Interval and symmetry properties can shorten the work

For integrable functions and constants α, β,

$$\int_a^c f = \int_a^b f + \int_b^c f, \qquad \int_a^b f = -\int_b^a f, \qquad \int_a^a f = 0,$$

$$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g.$$

Additivity uses signed orientation, so b need not lie between a and c when the convention $\int_u^v = -\int_v^u$ is retained.

Use symmetry only with its required interval or period

Hypothesis	Definite-integral consequence	Essential check
f is odd	integral on [-a,a] is zero	symmetric interval
f is even	integral on [-a,a] is twice 0 to a	symmetric interval
f has period T	any interval of length T gives same integral	full period length
reverse endpoints	integral changes sign	orientation
additivity	split at a convenient point	integrability on pieces

READ THE FIGURE

Odd/even symmetry concerns function values. Periodicity concerns translation of the whole interval. Neither is a substitute for checking convergence in an improper setting.

Odd powers cancel on symmetric intervals

For SII and Q25, x^3 is odd, so

$$\int_{-1}^{1} x^3 dx = 0.$$

Directly, $[x^4/4]_{-1}^{1} = 0$. For PII, $\sin^5 x$ is odd and the interval $[-\pi/2, \pi/2]$ is symmetric, hence its integral is also zero.

For the even function x^2 on the same type of interval,

$$\int_{-1}^{1} x^2\,dx = 2\int_{0}^{1} x^2\,dx = \tfrac{2}{3}.$$

Periodic intervals can be translated

If $f(x+T) = f(x)$ for every x, then for any a,

$$\int_{a}^{a+T} f(x)\,dx = \int_{0}^{T} f(x)\,dx.$$

For example, $|\sin x|$ has period π, so every interval of length π has the same total area 2.

COMMON TRAP

Cancellation is not positive area

The odd-function result gives signed cancellation. The total area of $y = x^3$ on $[-1, 1]$ is $2\int_{0}^{1} x^3\,dx = 1/2$, not zero. Use absolute values when the question asks for geometric area.

LESSON 06 / 17

Reflect the dummy variable across the interval

The substitution $x \mapsto a+b-x$ reverses orientation twice and gives

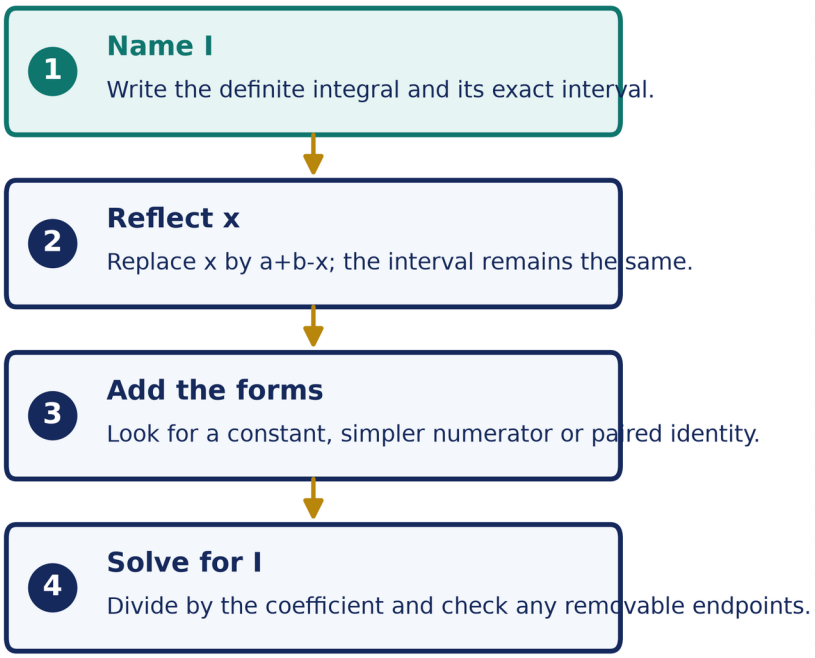
$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx.$$

On $[0,a]$, this is King's property

$$\int_0^a f(x)dx = \int_0^a f(a-x)dx.$$

It is most powerful when the original and reflected integrands can be added into something simpler.

King's property is a reflection followed by a useful combination



READ THE FIGURE
Reflection does not assert $f(x)=f(a+b-x)$. It asserts equality of the two integrals after substitution, even when the functions differ pointwise.

WORKED EXAMPLE

P10 and Q24: sine and cosine exchange places

Let $I = \int_0^{\pi/2} \sin^2 x\,dx$. Reflection gives $I = \int_0^{\pi/2} \cos^2 x\,dx$. Adding,

$$2I = \int_0^{\pi/2} (\sin^2 x + \cos^2 x)dx = \frac{\pi}{2},$$

so $I = \pi/4$.

WORKED EXAMPLE

S12 and Q26: add two reciprocal forms

For

$$I=\int_0^{\pi/2}\frac{dx}{1+\tan x},$$

reflection gives $I=\int_0^{\pi/2} dx/(1+\cot x)$. For $u>0$,

$$\frac{1}{1+u}+\frac{1}{1+1/u}=\frac{1}{1+u}+\frac{u}{1+u}=1.$$

Therefore $2I=\pi/2$ and $I=\pi/4$. At the endpoint $\pi/2$, the original expression is assigned its continuous limiting value 0.

Q36: the final answer is right, but the printed prose is not

For Q36,

$$\frac{\sin^2 x}{\sin^2 x+\cos^2 x}=\sin^2 x,$$

not $1/2$. Hence

$$\int_0^{\pi/2}\frac{\sin^2 x}{\sin^2 x+\cos^2 x}dx=\int_0^{\pi/2}\sin^2 x\,dx=\frac{\pi}{4}.$$

The source solution's final value is correct; its phrase "the integrand is $1/2$" is corrected here.

COMMON TRAP

Reflection does not remove an interior singularity

King's property preserves the interval and any singularities within it. If the integrand is improper, establish convergence and interpret the reflection through limits before adding the two forms.

LESSON 07 / 17

An infinite upper limit is always a limit of finite integrals

By definition,

$$\int_a^{\infty} f(x)dx = \lim_{R\to\infty} \int_a^{R} f(x)dx,$$

when that limit is finite. The symbol ∞ is not an endpoint at which an antiderivative can simply be evaluated without a limit.

WORKED EXAMPLE

P12 and Q27: an exponential tail has finite area

$$\int_0^{R} e^{-x}dx = 1 - e^{-R}.$$

As $R \to \infty$, $e^{-R} \to 0$, so

$$\int_0^{\infty} e^{-x}dx = 1.$$

The visual lab separates the finite part on $[0, 2]$ from the remaining tail of size e^{-2}.

The p-integrals give a fundamental comparison scale

For $p \neq 1$,

$$\int_1^{R} x^{-p}dx = \frac{R^{1-p} - 1}{1-p}.$$

Consequently, for $p > 1$,

$$\int_1^{\infty} \frac{dx}{x^p} = \frac{1}{p-1}.$$

For $p \leq 1$, the integral diverges. At $p = 1$, the finite value is $\ln R$, which diverges. The antiderivative formula and the limit must agree.

Infinite-interval convergence is decided by the tail

Integrand or test	Conclusion	Reason
1/x^p, p>1	converges to 1/(p-1)	negative exponent of R vanishes
1/x	diverges	ln R grows without bound
1/x^p, p<1	diverges	positive power of R grows
exp(-x)	converges to 1	exp(-R) vanishes
0<=f<=g	convergence of g forces f	comparison for nonnegative tails

READ THE FIGURE

Comparison tests require the stated sign and a common tail interval. A finite graph on a large finite window is not itself a proof of convergence.

WORKED EXAMPLE

Comparison can precede an exact evaluation

For $x \geq 1$, $0 < 1/(1+x^2) \leq 1/x^2$. Convergence follows by comparison, and FTC gives the exact value

$$\int_1^{\infty} \frac{dx}{1+x^2} = \lim_{R\to\infty} (\tan^{-1} R - \tan^{-1} 1) = \frac{\pi}{4}.$$

COMMON TRAP

A finite value at every R does not imply a finite limit

The number $\ln R$ is finite for each finite R, yet grows without bound. Always take the final limit. Also, signed cancellation over expanding intervals does not prove absolute convergence.

Approach a singular endpoint from inside the interval

If f is continuous on $(0,1]$ but unbounded near 0, define

$$\int_0^1 f(x)dx = \lim_{\varepsilon\to 0^+}\int_\varepsilon^1 f(x)dx$$

when the limit is finite. For $p<1$,

$$\int_0^1 x^{-p}dx = \frac{1}{1-p}.$$

The complete finite-interval p-test says that $\int_0^1 x^{-p}dx$ converges exactly when $p<1$; at $p=1$ it gives $-\ln\varepsilon\to\infty$.

WORKED EXAMPLE

P13: the logarithm is unbounded but integrable

Using $\int \ln x\,dx = x\ln x - x$,

$$\int_\varepsilon^1 \ln x\,dx = -1-(\varepsilon\ln\varepsilon-\varepsilon).$$

Since $\varepsilon\ln\varepsilon\to 0$ and $\varepsilon\to 0$,

$$\int_0^1 \ln x\,dx = -1.$$

WORKED EXAMPLE

S14/Q28 and P14/Q29: logarithmic powers still have finite area

For $x\ln x$,

$$\int_0^1 x\ln x\,dx = \lim_{\varepsilon\to 0^+}\left[\frac{x^2}{2}\ln x-\frac{x^2}{4}\right]_\varepsilon^1 = -\frac{1}{4}.$$

For $(\ln x)^2$,

$$\int_0^1 (\ln x)^2dx = \lim_{\varepsilon\to 0^+}\left[x((\ln x)^2-2\ln x+2)\right]_\varepsilon^1 = 2.$$

Both use $x(\ln x)^k\to 0$ as $x\to 0^+$ for fixed positive k.

Endpoint limits used by the logarithmic integrals

Limit as x->0+	Value	Why it matters
x ln x	0	removes the lower endpoint term
x (ln x)^2	0	permits the squared-log integral
x^2 ln x	0	permits the x ln x integral
x^p integral, p<1	finite	finite-interval p-test
ln x itself	-infinity	unbounded does not mean nonintegrable

READ THE FIGURE

The integrand may diverge while its signed or absolute area remains finite. Each endpoint term is evaluated as a limit, never by substituting x=0 into ln x.

COMMON TRAP

Distinguish an improper integral from a continuous removable expression

If an expression has a finite continuous extension, it can be assigned that value at the endpoint. If it remains unbounded, as $\ln x$ does, the integral must be defined through $\varepsilon \to 0^+$. In either case, the original domain and the limiting interpretation must be stated.

LESSON 09 / 17

Establish convergence before using a symmetry argument

For PIS and Q31, let

$$I=\int_0^{\infty}\frac{\ln x}{1+x^2}dx.$$

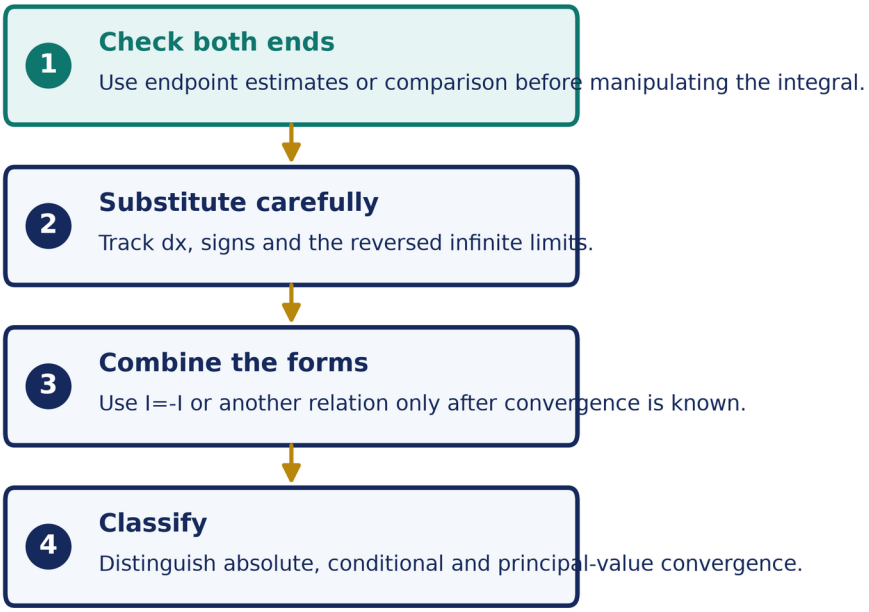
Near 0, $|\ln x|$ is integrable; near ∞, $|\ln x|/(1+x^2)=O(\ln x/x^2)$ is integrable. Thus the integral converges absolutely on both ends.

Now put $x=1/u$. Then $dx=-du/u^2$, $\ln x=-\ln u$, and the endpoints interchange. After simplifying,

$$I=\int_{\infty}^{0}\frac{\ln u}{1+u^2}du=-I.$$

Therefore $2I=0$ and $I=0$. The inversion is legitimate because convergence was checked first.

An improper symmetry argument has a convergence gate



READ THE FIGURE
For ln(x)/(1+x^2), inversion gives I=-I. The conclusion I=0 uses the fact that I is a finite number.

Three different meanings must not be confused

An integral $\int_I f$ converges **absolutely** when $\int_I |f|$ is finite. It converges **conditionally** when it converges but the absolute integral diverges. The Dirichlet integral in Extension VII is conditionally convergent.

A symmetric limit can define a Cauchy principal value without making the ordinary integral convergent. For example,

$$\int_{-R}^{R} \frac{x}{1+x^2}\,dx = 0$$

by oddness, yet

$$\int_{0}^{R} \frac{x}{1+x^2}\,dx = \frac{1}{2}\ln(1+R^2) \to \infty.$$

The two-sided improper integral therefore diverges even though its symmetric principal value is zero.

> **COMMON TRAP**
> **$I=-I$ is useful only for a finite I**
> If I were infinite or undefined, subtracting it from itself would be meaningless. Convergence is not a technical footnote; it is what permits the algebraic step.

LESSON 10 / 17

Geometric area uses nonnegative height

If $f(x) \geq 0$ on $[a, b]$, the area under $y = f(x)$ is $\int_a^b f(x)dx$. If f changes sign, total area is

$$\int_a^b |f(x)|dx,$$

computed by splitting at zeros. Between two curves with $f(x) \geq g(x)$ on an interval, vertical slices give

$$A = \int_a^b (f(x) - g(x))dx.$$

WORKED EXAMPLE

Signed cancellation and total area are different answers

For $y = x^3$ on $[-1, 1]$,

$$\int_{-1}^{1} x^3 dx = 0,$$

but

$$A = \int_{-1}^{0} (-x^3)dx + \int_0^1 x^3 dx = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}.$$

The visual lab shades the two positive geometric pieces separately.

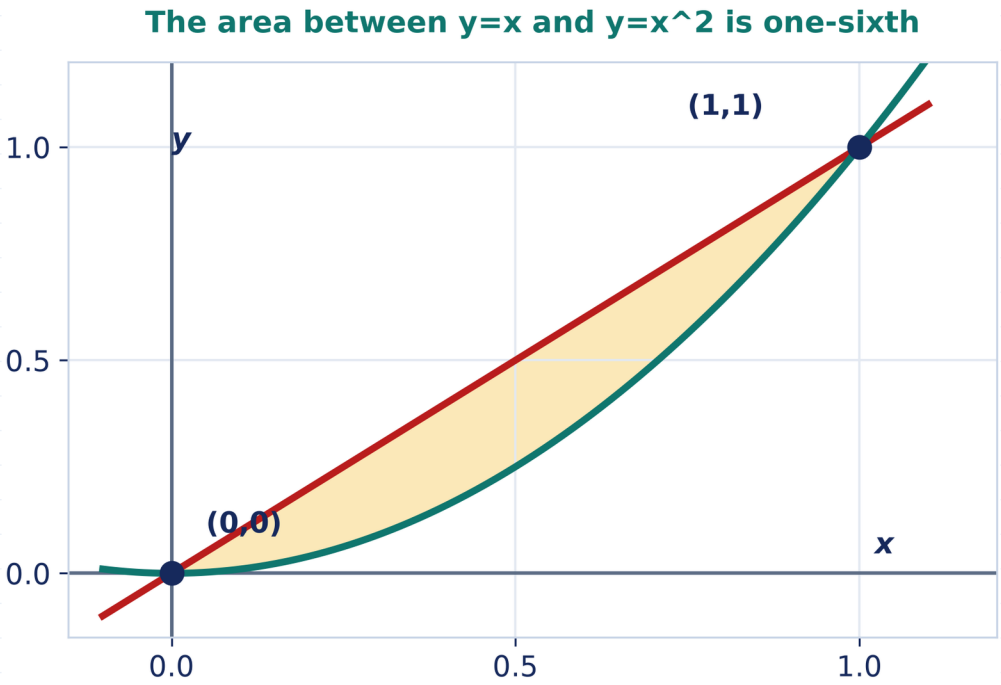
WORKED EXAMPLE

Area between a line and a parabola

On $[0, 1]$, $x \geq x^2$. Hence

$$A = \int_0^1 (x - x^2)dx = \left[\tfrac{x^2}{2} - \tfrac{x^3}{3}\right]_0^1 = \tfrac{1}{6}.$$

The intersections at $x = 0, 1$ supply the finite boundary of the shaded region.



READ THE FIGURE

The red line is the upper boundary and the teal parabola is the lower boundary on [0,1]. The gold height is x-x^2, whose integral is 1/6.

A familiar arch confirms the sign convention

For P9, $\sin x \geq 0$ on $[0, \pi]$, so its integral and geometric area agree:

$$\int_0^{\pi} \sin x \, dx = [-\cos x]_0^{\pi} = 2.$$

On an interval where the sine is negative, the signed integral would be negative while total area would use $-\sin x$.

COMMON TRAP

Find intersections before choosing upper and lower curves

The ordering $f \geq g$ can change at an intersection. A single formula $f - g$ over the whole interval would then subtract the two regions instead of adding their positive areas.

LESSON 11 / 17

Split when the upper and lower curves exchange places

Between $y=\sin x$ and $y=\cos x$ on $[0,\pi/2]$, the intersection is $x=\pi/4$. Therefore

$$A=\int_0^{\pi/4}(\cos x-\sin x)dx+\int_{\pi/4}^{\pi/2}(\sin x-\cos x)dx.$$

Each piece equals $\sqrt{2}-1$, so

$$A=2\sqrt{2}-2.$$

Visual lab sheet 6 shades both positive pieces.

Horizontal slices can avoid splitting

For the region between $x=y^2$ and $x=y+2$, solve

$$y^2=y+2 \quad\Longrightarrow\quad y=-1,2.$$

For $-1\le y\le 2$, the line is to the right: $y+2\ge y^2$. Thus

$$A=\int_{-1}^{2}(y+2-y^2)dy=\left[\frac{y^2}{2}+2y-\frac{y^3}{3}\right]_{-1}^{2}=\frac{9}{2}.$$

Here the strip height is dy and its horizontal length is right-minus-left.

An ellipse is a stretched circle

For $x^2/a^2+y^2/b^2=1$ with $a,b>0$,

$$y=\pm b\sqrt{1-\frac{x^2}{a^2}}.$$

Hence

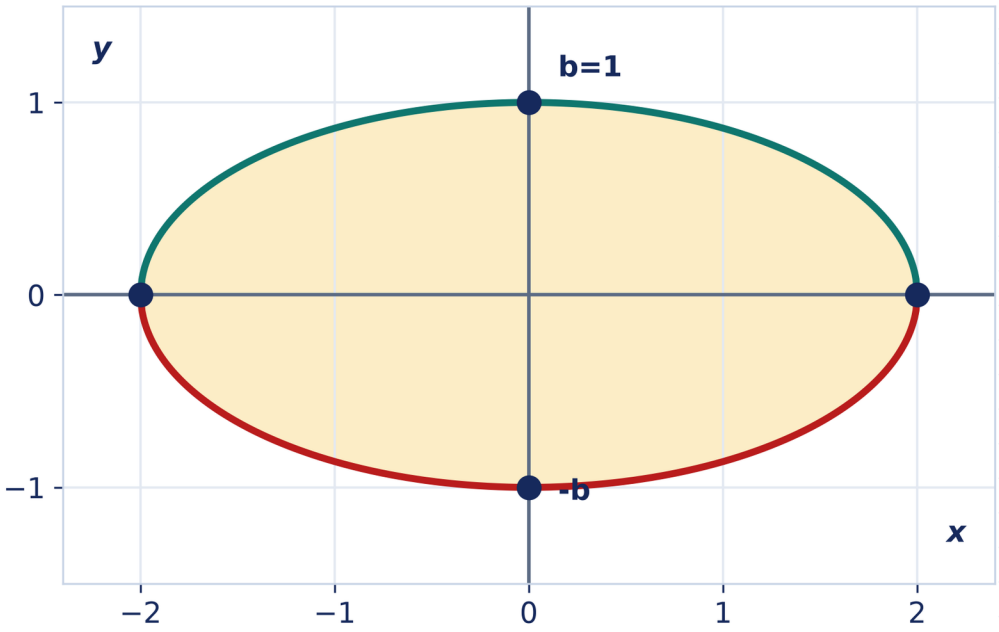
$$A=2b\int_{-a}^{a}\sqrt{1-\frac{x^2}{a^2}}dx.$$

With $x=au$, the integral becomes $2ab\int_{-1}^{1}\sqrt{1-u^2}\,du=2ab(\pi/2)$, so

$$A=\pi ab.$$

This also gives the circle area πa^2 when $b=a$.

Vertical slices of an ellipse give area pi ab



READ THE FIGURE

The ellipse has a=2 and b=1. Its vertical height is 2b sqrt(1-x^2/a^2), and the shaded area is pi ab=2pi.

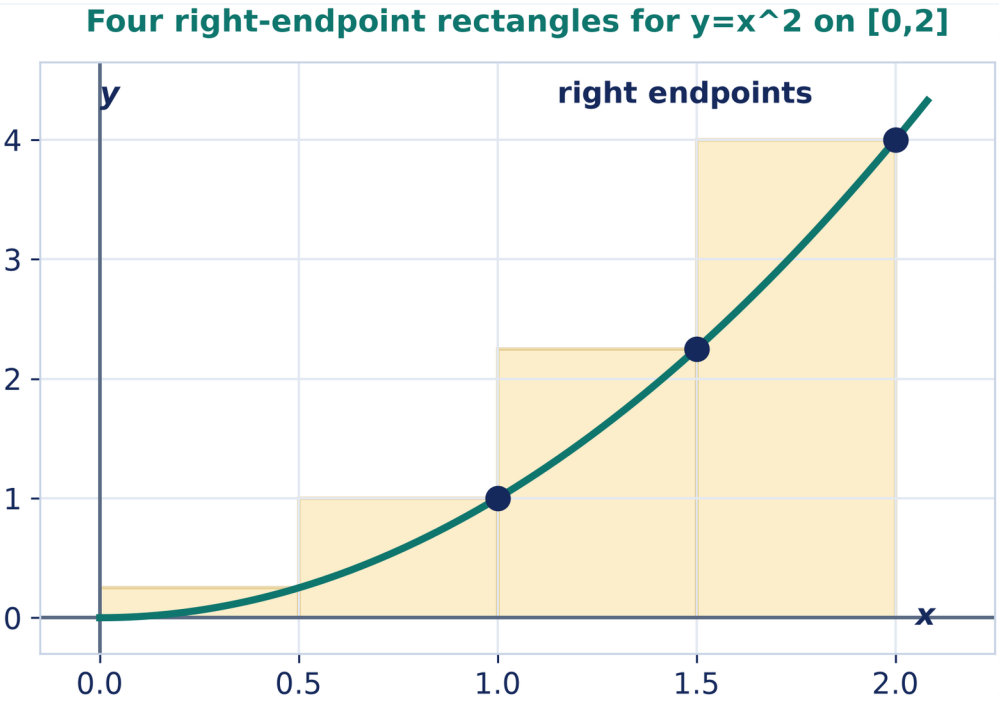
COMMON TRAP

Choose the variable that matches the boundary equations

Use top-minus-bottom with dx and right-minus-left with dy. Do not insert x-equations into a dy integral without first expressing both boundaries as functions of y.

LESSON 12 / 17

Read the approximation before taking the limit



READ THE FIGURE

The four heights are 1/4, 1, 9/4 and 4, with common width 1/2. Their sum is 3.75, above the exact integral 8/3 because x^2 is increasing.

Facts to retain

- A right-endpoint sum uses $f(x_i)$ with $x_i = a + i\Delta x$.
- For an increasing nonnegative function, right-endpoint rectangles overestimate the area; left endpoints underestimate it.
- Refinement changes both the number and widths of rectangles, not the target interval.

WORKED EXAMPLE

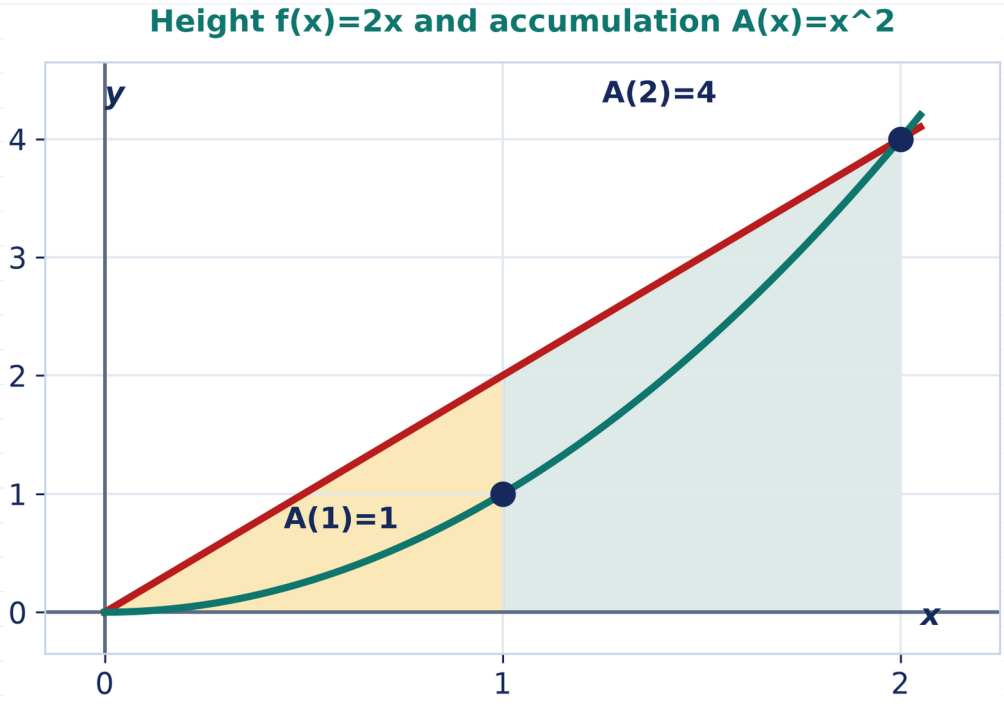
Check the displayed finite sum

$$R_4 = \tfrac{1}{2}\left(\tfrac{1}{4} + 1 + \tfrac{9}{4} + 4\right) = \tfrac{15}{4} = 3.75.$$

As $n \to \infty$, $R_n \to 8/3 \approx 2.6666667$. The finite sum and its limit are both useful, but they are not equal.

LESSON 13 / 17

See FTC as a moving upper boundary



READ THE FIGURE

The red line is the height 2x; the teal parabola is A(x)=integral from 0 to x of 2t dt=x^2. Gold and pale teal areas accumulate to 1 and 4.

Facts to retain

- $A(x) = \int_0^x 2t\,dt = x^2$, while $A'(x) = 2x = f(x)$.
- The area added from x to $x + h$ is approximately $f(x)h$ for small h.
- The derivative of the area curve is the height curve, not the area under the height curve again.

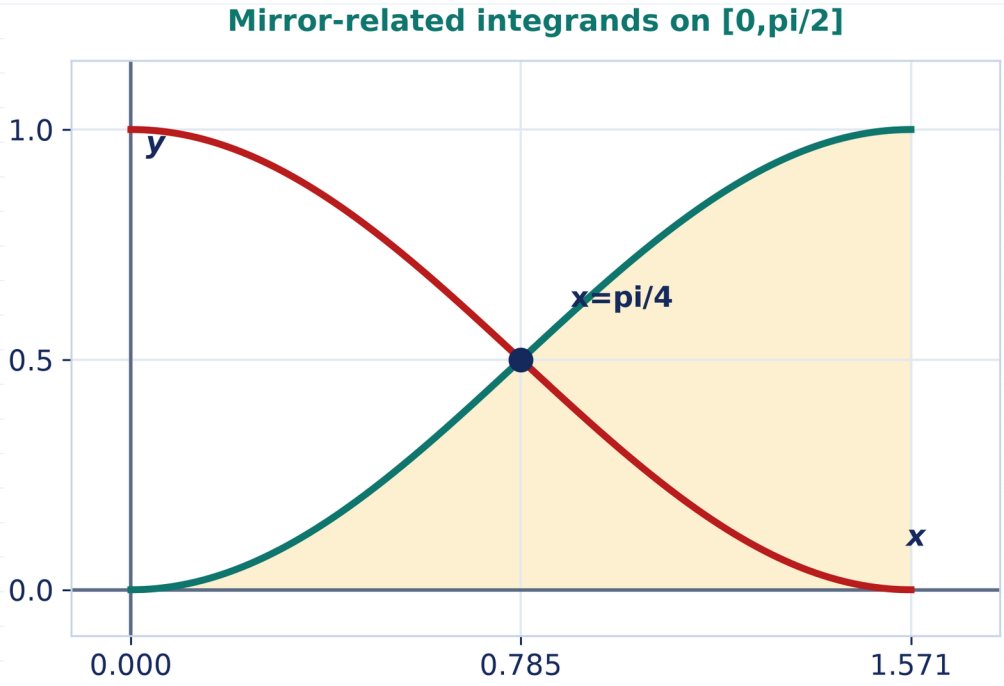
WORKED EXAMPLE

Verify both directions numerically

At $x = 1$, the triangular gold area is $\frac{1}{2}(1)(2) = 1 = A(1)$. Also $A'(1) = 2 = f(1)$. At $x = 2$, the total triangular area is $\frac{1}{2}(2)(4) = 4 = A(2)$.

LESSON 14 / 17

Reflection swaps sine squared and cosine squared



READ THE FIGURE

Teal sin^2(x) and red cos^2(x) interchange under x->pi/2-x. Each has area pi/4, and their ordinates add to 1 at every x.

Facts to retain

- Reflection preserves the interval $[0, \pi/2]$ and the integral, not each individual height.
- At every x, $\sin^2 x + \cos^2 x = 1$; hence the two equal areas sum to the rectangle area $\pi/2$.
- The crossing at $x = \pi/4$ has common height $1/2$.

WORKED EXAMPLE

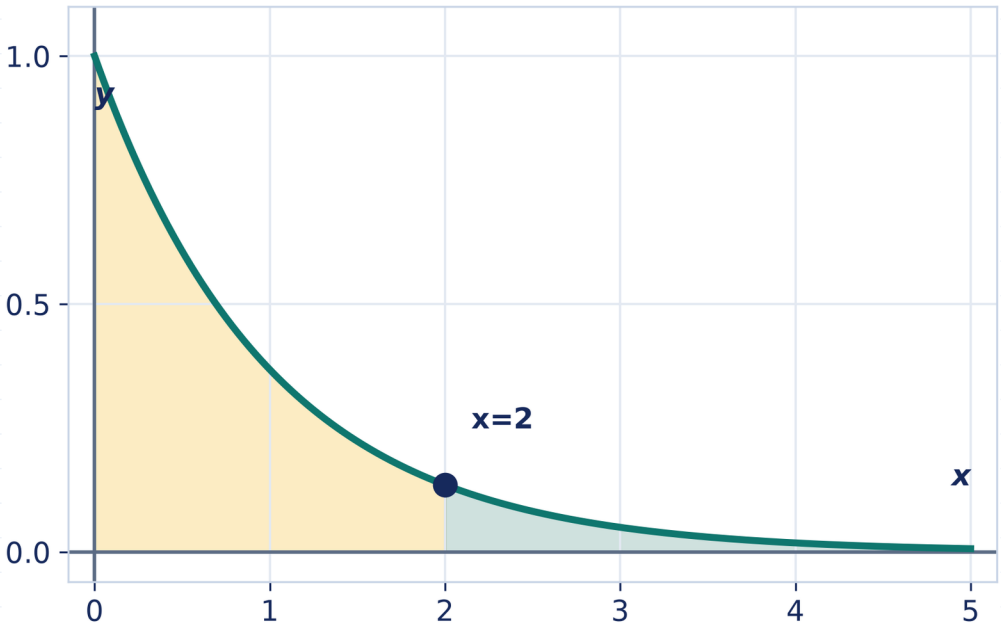
Read the two areas from the diagram

If I is either shaded area, then $2I = \int_0^{\pi/2} 1\,dx = \pi/2$, so $I = \pi/4 \approx 0.7853981634$. The vertical midpoint $\pi/4$ and the area value $\pi/4$ are numerically equal here only because the interval length is $\pi/2$.

LESSON 15 / 17

The tail continues forever but shrinks exponentially

The exponential area on [0,infinity) equals one



READ THE FIGURE

The gold area on [0,2] is 1-exp(-2)=0.8646647. The entire tail beyond 2 has area exp(-2)=0.1353353; only its portion up to 5 is visible.

Facts to retain

- $\int_0^2 e^{-x}dx = 1 - e^{-2}$, while $\int_2^\infty e^{-x}dx = e^{-2}$.
- The two contributions add to 1, exactly matching $\int_0^\infty e^{-x}dx$.
- A finite plotting window illustrates a tail but does not replace the limit $R \to \infty$.

WORKED EXAMPLE

Check tail additivity

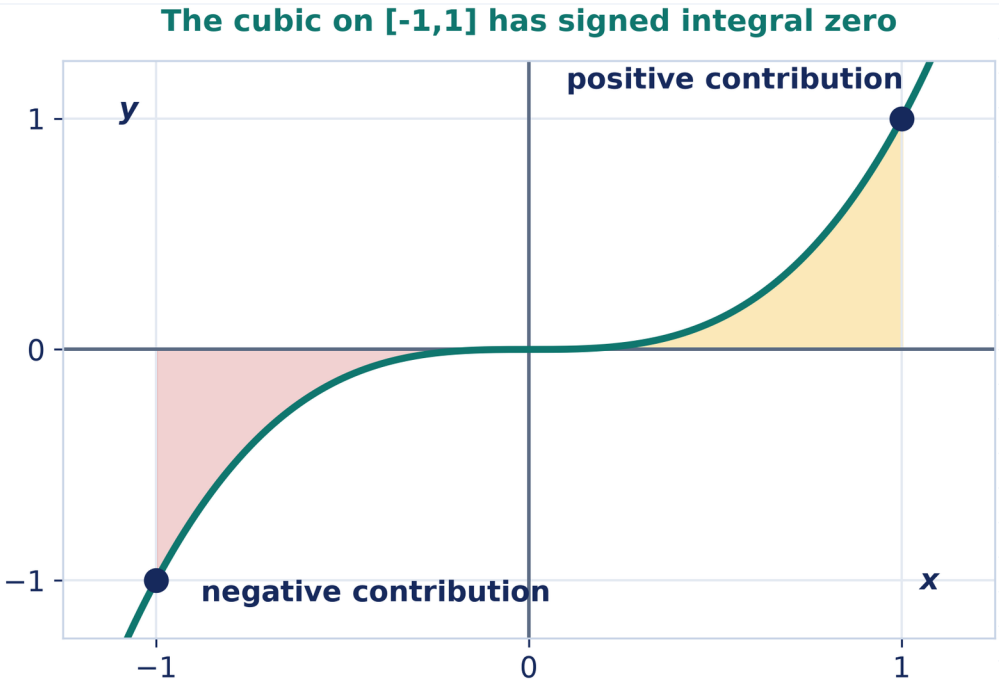
For any $R > 2$,

$$\int_0^2 e^{-x}dx + \int_2^R e^{-x}dx = 1 - e^{-R}.$$

Taking $R \to \infty$ gives $(1 - e^{-2}) + e^{-2} = 1$.

LESSON 16 / 17

Opposite signs cancel; geometric areas add



READ THE FIGURE

The red and gold geometric areas are each 1/4. Their signed contributions are -1/4 and +1/4, so the definite integral is 0 while total area is 1/2.

Facts to retain

- Oddness on a symmetric interval explains signed cancellation immediately.
- Total area uses $|x^3|$ and is twice the positive piece on $[0, 1]$.
- The zero at $x = 0$ is the required splitting point for the absolute-value calculation.

WORKED EXAMPLE

Compute both quantities without confusing them

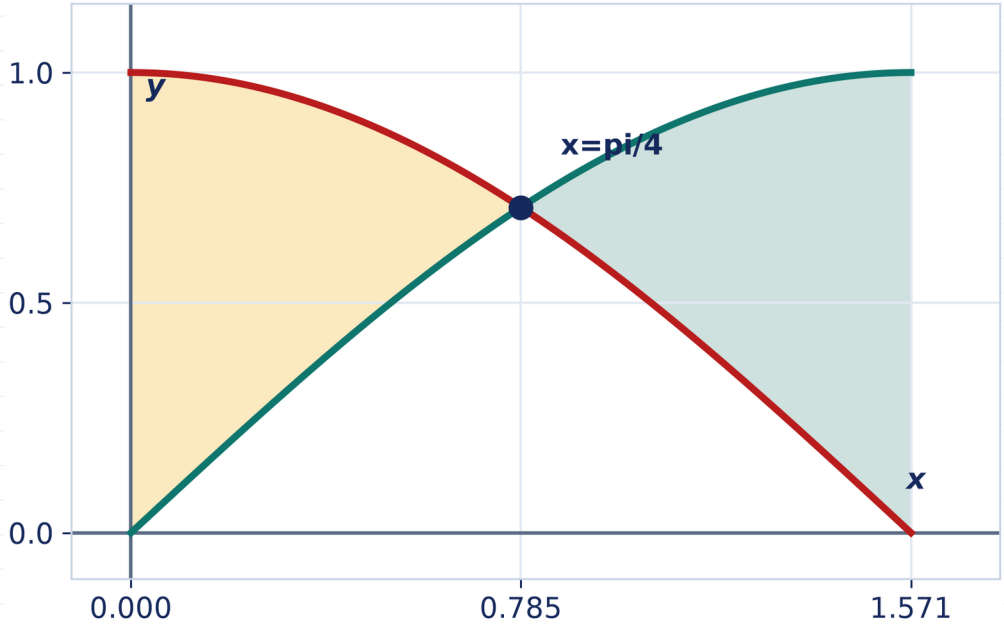
$$\int_{-1}^{1} x^3\,dx = 0, \qquad \int_{-1}^{1} |x^3|\,dx = 2\int_{0}^{1} x^3\,dx = \tfrac{1}{2}.$$

The first is an oriented algebraic quantity; the second is the shaded geometric area.

LESSON 17 / 17

The top curve changes at the intersection

Two positive regions between sine and cosine



READ THE FIGURE

Cosine is above sine before pi/4; sine is above cosine afterwards. The gold and pale teal areas are each sqrt(2)-1, giving total area 2sqrt(2)-2.

Facts to retain

- The intersection satisfies $\sin x = \cos x$, hence $x = \pi/4$ on the displayed interval.
- Using one fixed top-minus-bottom formula across $\pi/4$ would produce signed cancellation instead of total area.
- Each positive piece has area $\sqrt{2} - 1 \approx 0.4142135624$.

WORKED EXAMPLE

Check the two piecewise antiderivatives

$$\int_0^{\pi/4} (\cos x - \sin x)\,dx = [\sin x + \cos x]_0^{\pi/4} = \sqrt{2} - 1,$$

$$\int_{\pi/4}^{\pi/2} (\sin x - \cos x)\,dx = [-\cos x - \sin x]_{\pi/4}^{\pi/2} = \sqrt{2} - 1.$$

LESSON 01 / 07

A recurrence converts powers into two base cases

For $n \geq 0$, define

$$I_n = \int_0^{\pi/2} \sin^n x \, dx.$$

For $n \geq 2$, integration by parts gives

$$I_n = (n-1)\int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx = (n-1)(I_{n-2} - I_n),$$

so

$$I_n = \tfrac{n-1}{n} I_{n-2}, \qquad I_0 = \tfrac{\pi}{2}, \qquad I_1 = 1.$$

Consequently,

$$I_{2m} = \tfrac{(2m-1)!!}{(2m)!!}\tfrac{\pi}{2}, \qquad I_{2m+1} = \tfrac{(2m)!!}{(2m+1)!!}.$$

WORKED EXAMPLE

An odd-power check, followed by SIS and Q30

For the additional odd-power check $\sin^5 x$,

$$I_5 = \tfrac{4}{5} I_3 = \tfrac{4}{5} \cdot \tfrac{2}{3} I_1 = \tfrac{8}{15}.$$

For SIS and Q30, the requested power is $\sin^6 x$:

$$I_6 = \tfrac{5}{6} \cdot \tfrac{3}{4} \cdot \tfrac{1}{2} \cdot \tfrac{\pi}{2} = \tfrac{5\pi}{32}.$$

Wallis reduction separates even and odd powers

Power or index	Reduction route	Exact value
n=0	base integral of 1	pi/2
n=1	base integral of sin x	1
n=5	(4/5)(2/3) I1	8/15
n=6	(5/6)(3/4)(1/2) I0	5 pi/32
even 2m	double-factorial form times pi/2	decreasing in m
odd 2m+1	double-factorial form	decreasing in m

READ THE FIGURE

The recurrence is exact for every n>=2. It also supplies the ratios needed to prove the infinite Wallis product rather than merely quoting it.

The infinite product follows by squeezing adjacent integrals

On $[0, \pi/2]$, $0 \le \sin x \le 1$, hence

$$I_{2m+1} \le I_{2m} \le I_{2m-1}.$$

The recurrence gives $I_{2m-1}/I_{2m+1} = (2m+1)/(2m) \to 1$, so $I_{2m}/I_{2m+1} \to 1$. Substituting the closed forms and rearranging yields Wallis's product:

$$\frac{\pi}{2} = \lim_{m\to\infty} \frac{2}{1}\frac{2}{3}\frac{4}{3}\frac{4}{5}\cdots\frac{2m}{2m-1}\frac{2m}{2m+1}.$$

This is a convergent infinite product, not an assertion that any finite partial product equals $\pi/2$.

COMMON TRAP

Double factorials skip every other factor

For example, $6!! = 6\cdot 4\cdot 2$ and $5!! = 5\cdot 3\cdot 1$. They do not mean repeated factorial operations. Always retain the base factor $\pi/2$ for even powers and omit it for odd powers.

LESSON 02 / 07

Two convergent parameter integrals and their bridge

For $z>0$, the Gamma function is

$$\Gamma(z)=\int_0^\infty t^{z-1}e^{-t}dt, \qquad \Gamma(z+1)=z\Gamma(z),$$

with $\Gamma(1)=1$ and $\Gamma(n+1)=n!$. For $p,q>0$,

$$B(p,q)=\int_0^1 t^{p-1}(1-t)^{q-1}dt=\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

The product identity follows from a double integral and a factorising change of variables. The substitution $t=\sin^2\theta$ gives

$$B(p,q)=2\int_0^{\pi/2}\sin^{2p-1}\theta\cos^{2q-1}\theta\,d\theta.$$

WORKED EXAMPLE

S18 and Q39: the base Beta value

$$B\left(\tfrac12,\tfrac12\right)=2\int_0^{\pi/2}1\,d\theta=\pi=\frac{\Gamma(1/2)^2}{\Gamma(1)}.$$

Since $\Gamma(1/2)>0$, this also gives $\Gamma(1/2)=\sqrt{\pi}$.

The same Beta formula evaluates Q40 and S19/Q43

For $p>-1$, putting $q=1/2$ in the trigonometric Beta form gives

$$\int_0^{\pi/2}\sin^p x\,dx=\tfrac12 B\left(\frac{p+1}{2},\tfrac12\right)=\frac{\sqrt{\pi}}{2}\frac{\Gamma((p+1)/2)}{\Gamma((p+2)/2)}.$$

For Q40, first use $\sin x\cos x=\sin(2x)/2$. With $u=2x$ and sine symmetry on $[0,\pi]$,

$$A=\frac{1}{\sqrt2}\int_0^{\pi/2}\sqrt{\sin(2x)}\,dx=\frac{1}{2\sqrt2}\int_0^{\pi}\sqrt{\sin u}\,du=\frac{1}{\sqrt2}\int_0^{\pi/2}\sqrt{\sin u}\,du.$$

Now put $p=1/2$ in the Beta formula:

$$A=\frac{1}{\sqrt2}\frac{\sqrt{\pi}}{2}\frac{\Gamma(3/4)}{\Gamma(5/4)}\approx 0.8472130848.$$

For S19/Q43, put $p=6$:

$$\int_0^{\pi/2}\sin^6 x\,dx=\frac{\sqrt{\pi}}{2}\frac{\Gamma(7/2)}{\Gamma(4)}=\frac{5\pi}{32},$$

using $\Gamma(7/2)=15\sqrt{\pi}/8$ and $\Gamma(4)=6$.

Q42 and Q44: reflection and duplication

For $0 < z < 1$ in the real-integral setting, Euler's reflection formula is

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Analytic continuation extends it to nonintegral complex values and explains Gamma's poles at the nonpositive integers.

Legendre's duplication formula, for $z > 0$, is

$$\Gamma(z)\Gamma\left(z+\tfrac{1}{2}\right) = 2^{1-2z}\sqrt{\pi}\,\Gamma(2z).$$

At $z = 1/2$, it reduces to $\Gamma(1/2) = \sqrt{\pi}$, not $\pi = 2\sqrt{\pi}$; the source's substitution at that point is corrected. For P18/Q41, $z = 1/4$ gives

$$\Gamma\left(\tfrac{1}{4}\right)\Gamma\left(\tfrac{3}{4}\right) = 2^{1/2}\sqrt{\pi}\,\Gamma\left(\tfrac{1}{2}\right) = \sqrt{2\pi},$$

in agreement with Q42's reflection formula.

Parameter identities must be checked at the substituted value

Source item	Parameter choice	Conclusion
S18 / Q39	p=q=1/2	B=pi and Gamma(1/2)=sqrt(pi)
Q40	p=1/2 after u=2x	0.8472130848 approximately
S19 / Q43	p=6	5 pi/32
Q42 / Q44	reflection and duplication	identities with stated domains
P18 / Q41	z=1/4	product=sqrt(2) pi

READ THE FIGURE

The z=1/2 duplication step is only a consistency check. The z=1/4 evaluation is the nontrivial product and agrees with reflection.

COMMON TRAP

Parameter restrictions are convergence restrictions

The stated real-integral definitions require $z > 0$ and $p, q > 0$. Analytic continuation extends Gamma to other complex values, but it does not make the original real integral converge there.

LESSON 03 / 07

S16/Q32: polar coordinates square a one-dimensional integral

Let

$$I = \int_0^\infty e^{-x^2} dx.$$

The exponential tail proves convergence, and $I > 0$. Therefore

$$I^2 = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx\, dy.$$

On the first quadrant, polar coordinates give $x^2 + y^2 = r^2$ and Jacobian r:

$$I^2 = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r\, dr\, d\theta = \tfrac{\pi}{2} \cdot \tfrac{1}{2} = \tfrac{\pi}{4}.$$

Thus

$$\int_0^\infty e^{-x^2} dx = \tfrac{\sqrt{\pi}}{2}, \qquad \int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}.$$

P16/Q33 and Q46: scale, shift and complete the square

For $a > 0$, $u = \sqrt{a}\,x$ gives

$$\int_{-\infty}^\infty e^{-ax^2} dx = \tfrac{1}{\sqrt{a}} \int_{-\infty}^\infty e^{-u^2} du = \sqrt{\tfrac{\pi}{a}}.$$

P16/Q33 is the case $a = 2$, with value $\sqrt{\pi/2}$.

For Q46, also complete the square:

$$-ax^2 + bx = -a\left(x - \tfrac{b}{2a}\right)^2 + \tfrac{b^2}{4a}.$$

Translation does not change a full-line integral, so

$$\int_{-\infty}^\infty e^{-ax^2 + bx} dx = e^{b^2/(4a)} \sqrt{\tfrac{\pi}{a}}, \qquad a > 0.$$

The exponential factor comes from the completed-square constant; the factor $\sqrt{\pi/a}$ comes from scaling.

Additional extension: differentiate a cosine parameter

Define

$$J(a) = \int_0^\infty e^{-x^2} \cos(2ax) dx.$$

Differentiation under the integral is justified by the integrable dominating function $2xe^{-x^2}$. Hence

$$J'(a) = -\int_0^\infty 2xe^{-x^2}\sin(2ax)dx.$$

Integration by parts with $d(e^{-x^2}) = -2xe^{-x^2}dx$ gives

$$J'(a) = -2aJ(a), \qquad J(0) = \frac{\sqrt{\pi}}{2}.$$

Solving yields $J(a) = J(0)e^{-a^2}$. In particular, $J(1) = \sqrt{\pi}/(2e)$, and the full-line cosine transform is $\sqrt{\pi}e^{-a^2}$.

The Gaussian and its transform are positive, exact and checkable

Integral	Method	Value
S16 / Q32 half-line	square and use polar coordinates	sqrt(pi)/2
P16 / Q33 with a=2	scale x by sqrt(a)	sqrt(pi/2)
Q46 with linear term	complete square, translate and scale	sqrt(pi/a) exp(b^2/(4a))
additional cosine transform	differentiate then integrate by parts	sqrt(pi) exp(-a^2)/2

READ THE FIGURE

The three canonical Gaussian results all follow from the same half-line value. The cosine transform is an additional parameter-differentiation extension.

COMMON TRAP

The Jacobian and scaling factors are essential

Replacing $dx\,dy$ by $dr\,d\theta$ without r would give the wrong polar value. Likewise, $u = \sqrt{a}x$ contributes $1/\sqrt{a}$. Also, $I^2 = \pi/4$ determines $|I|$; positivity selects $I = \sqrt{\pi}/2$.

LESSON 04 / 07

Scaling two copies of the same function

Suppose f is continuous with finite limiting values $f(0)$ and $f(\infty)$. For $a, b > 0$, Frullani's formula is

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(\infty)) \ln \frac{b}{a}.$$

Indeed, after $u = ax$ and $u = bx$ in the two pieces of a finite cutoff integral,

$$\int_\varepsilon^R \frac{f(ax) - f(bx)}{x} dx = \int_{a\varepsilon}^{b\varepsilon} \frac{f(u)}{u} du - \int_{aR}^{bR} \frac{f(u)}{u} du.$$

The endpoint limiting values produce the stated logarithm.

WORKED EXAMPLE

S17/Q34 and P17/Q38: exponential Frullani integrals

For $f(t) = e^{-t}$, $f(0) = 1$ and $f(\infty) = 0$. Therefore

$$\int_0^\infty \frac{e^{-x} - e^{-2x}}{x} dx = \ln 2,$$

and

$$\int_0^\infty \frac{e^{-3x} - e^{-5x}}{x} dx = \ln \frac{5}{3} \approx 0.5108256238.$$

More generally, for $x > 0$, $\ln x = \int_0^\infty (e^{-t} - e^{-xt})/t \, dt$.

Additional conditional example: a cosine difference

For $a, b > 0$, $\cos(ax)$ has no limit at infinity, so the basic finite-$f(\infty)$ statement does not directly apply. Introduce $\delta > 0$ and use

$$\frac{\cos(ax) - \cos(bx)}{x} = \int_a^b \sin(tx) dt.$$

Fubini's theorem for the damped integral gives

$$I_\delta = \int_a^b \frac{t}{\delta^2 + t^2} dt = \frac{1}{2} \ln \frac{\delta^2 + b^2}{\delta^2 + a^2}.$$

Dirichlet convergence and Abel continuity as $\delta \to 0^+$ then give

$$\int_0^\infty \frac{\cos(ax) - \cos(bx)}{x} dx = \ln \frac{b}{a}.$$

Source-warning correction: the sine difference converges to zero

For the additional sine-difference example,

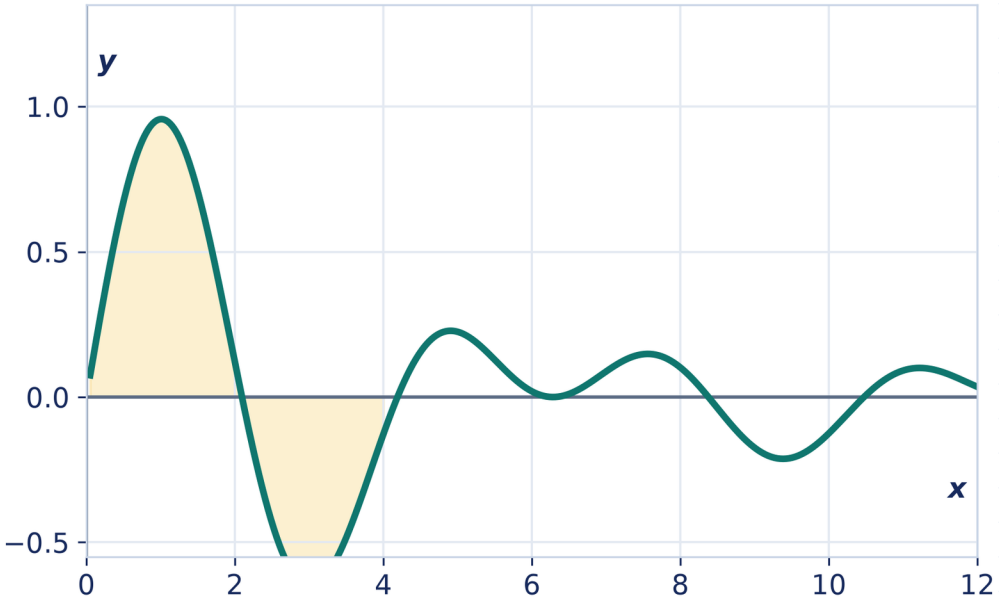
$$\frac{\sin(ax)-\sin(bx)}{x} = -\int_a^b \cos(tx)\,dt.$$

With the same damping,

$$I_\delta = -\int_a^b \frac{\delta}{\delta^2+t^2}\,dt = -\left[\tan^{-1}\frac{t}{\delta}\right]_a^b \longrightarrow 0.$$

Dirichlet's test proves convergence of the original improper integral, and Abel continuity identifies its value as 0. The source's claim that this difference diverges merely because sine has no limiting value at infinity is incorrect.

A cosine Frullani integrand with a1 and b2



READ THE FIGURE
For a=1 and b=2, the full conditional integral is ln(2). The shaded finite window is illustrative; convergence and the exact value come from the damped calculation, not the plot.

COMMON TRAP
No f(infinity) does not automatically mean divergence
The finite-limit Frullani theorem has hypotheses, but failure of one shortcut does not settle convergence. The exponential source questions satisfy the hypotheses directly; oscillatory differences require a separate convergence argument and can converge conditionally.

LESSON 05 / 07

Differentiate a parameter, then recover the constant

Under suitable local domination and convergence hypotheses,

$$I(a)=\int f(x,a)\,dx \quad\Longrightarrow\quad I'(a)=\int \frac{\partial f}{\partial a}(x,a)\,dx.$$

After evaluating the simpler derivative, integrate in a and use one known value to determine the constant.

WORKED EXAMPLE

S14/Q28 and P14/Q29: differentiate a power integral

For $a>-1$,

$$F(a)=\int_0^1 x^a\,dx=\frac{1}{a+1}.$$

Differentiating once and twice gives

$$\int_0^1 x^a\ln x\,dx=-\frac{1}{(a+1)^2},\qquad \int_0^1 x^a(\ln x)^2\,dx=\frac{2}{(a+1)^3}.$$

At $a=1$, the first identity gives S14/Q28 $=-1/4$. At $a=0$, the second gives P14/Q29 $=2$. These agree with the by-parts evaluations in Lesson 8.

Q45: a logarithmic quotient represented by a parameter integral

For $a>-1$, define

$$I(a)=\int_0^1 \frac{x^a-1}{\ln x}\,dx.$$

At $x=1$, the expression has continuous limiting value a; convergence at 0 follows from the x^a scale and the logarithmic denominator. Differentiation gives

$$I'(a)=\int_0^1 x^a\,dx=\frac{1}{a+1}.$$

Since $I(0)=0$,

$$I(a)=\int_0^a \frac{dt}{1+t}=\ln(1+a).$$

Thus Q45, with $a=3$, equals $\ln 4=2\ln 2$.

The two anchors are consistent: $I(0)=0=\ln 1$, and $I'(a)=1/(1+a)$ is exactly the derivative of $\ln(1+a)$. For $-1<a<0$, the oriented parameter integral and the logarithm are both negative, providing an additional sign check.

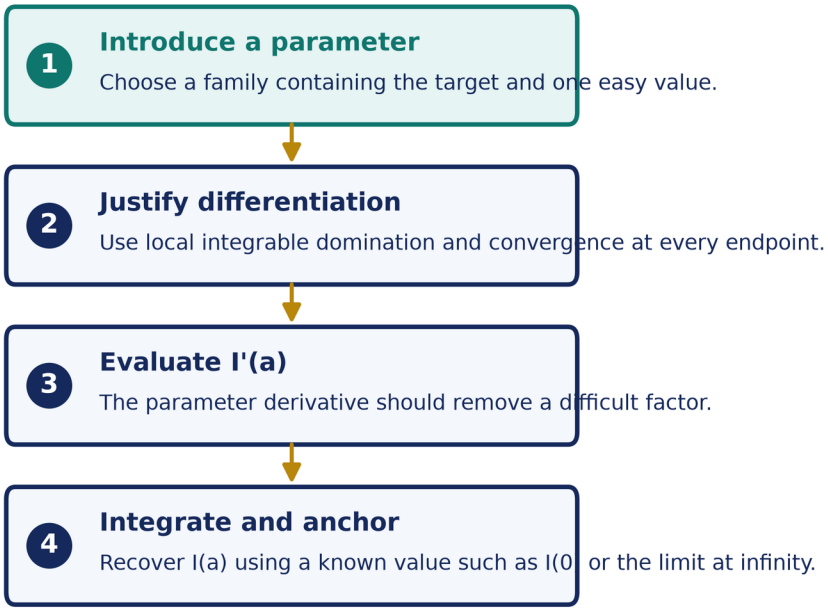
Q45's source representation needs a corrected lower limit

For $x > 0$, $x \neq 1$,

$$\int_0^a x^t dt = \left[\frac{x^t}{\ln x}\right]_0^a = \frac{x^a - 1}{\ln x}.$$

If the lower limit were 1 as printed in the source, the value would instead be $(x^a - x)/\ln x$. Thus the intended representation requires lower limit 0; the final Q45 answer $\ln 4$ remains correct.

Feynman differentiation is a four-step method with hypotheses



READ THE FIGURE
For Q45, the anchor is $I(0)=0$. For the power derivatives, differentiation gives the answer directly without an integration constant.

COMMON TRAP
Differentiation under the integral sign needs its own hypotheses
Convergence of the original integral alone is not sufficient. Use local integrable domination for the parameter derivative, check improper endpoints, and anchor the integrated result with a known value.

LESSON 06 / 07

First evaluate the damped parameter family

For $t > 0$, define

$$J(t) = \int_0^\infty e^{-tx}\frac{\sin x}{x}dx,$$

with continuous value 1 for $\sin x/x$ at $x = 0$. Local domination permits differentiation:

$$J'(t) = -\int_0^\infty e^{-tx}\sin x\,dx = -\frac{1}{1+t^2}.$$

Since $J(t) \to 0$ as $t \to \infty$,

$$J(t) = \int_t^\infty \frac{ds}{1+s^2} = \frac{\pi}{2} - \tan^{-1} t.$$

This proves the damped formula without setting $t = 0$ prematurely.

Convergence comes from oscillation and a decreasing envelope

The expression $\sin x/x$ has continuous extension 1 at $x = 0$, so only infinity is improper. For $1 < R < S$, integration by parts gives

$$\int_R^S \frac{\sin x}{x}dx = \left[-\frac{\cos x}{x}\right]_R^S - \int_R^S \frac{\cos x}{x^2}dx.$$

The boundary terms are $O(1/R)$ and the remaining absolute integral is at most $\int_R^\infty x^{-2}dx = 1/R$. Thus the finite integrals have a limit.

The convergence is not absolute. On each interval $[n\pi + \pi/6, n\pi + 5\pi/6]$, $|\sin x| \geq 1/2$, so the contribution to $\int |\sin x|/x\,dx$ is at least a constant multiple of $1/(n+1)$. The harmonic series diverges.

Abel continuity supplies the exact value

For $\delta > 0$, the calculation above gives

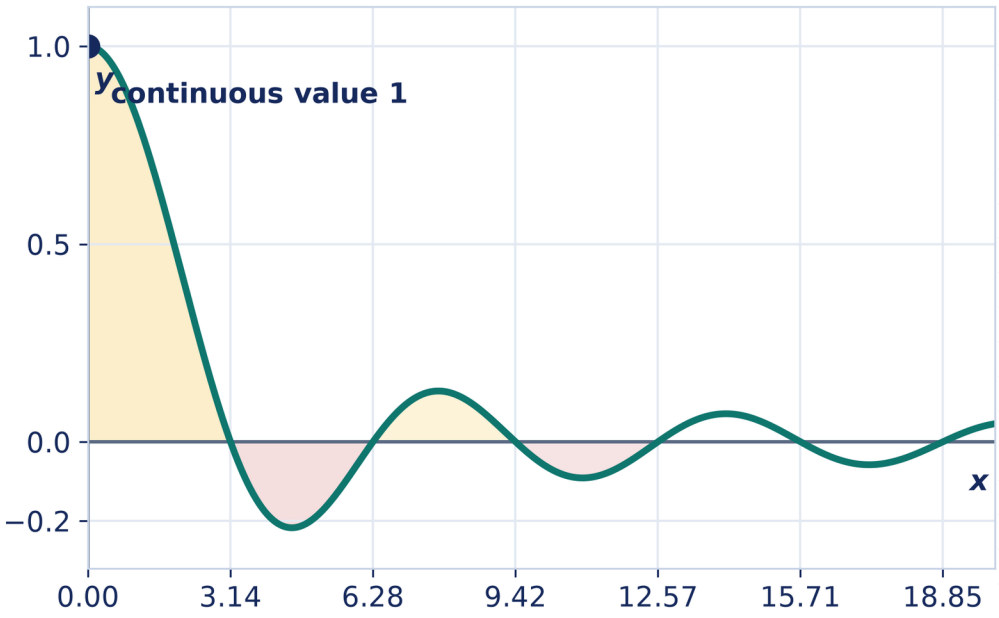
$$I(\delta) = \int_0^\infty e^{-\delta x}\frac{\sin x}{x}dx = \frac{\pi}{2} - \tan^{-1}\delta.$$

Because the undamped improper integral converges, Abel continuity gives Q37:

$$\int_0^\infty \frac{\sin x}{x}dx = \lim_{\delta \to 0^+} I(\delta) = \frac{\pi}{2}.$$

This is the missing justification for inserting a convergence factor and then removing it.

Alternating Dirichlet arches have a finite signed sum



READ THE FIGURE

Alternating arches make the improper integral converge conditionally to pi/2. Their absolute areas do not have a finite sum, and the graph cannot by itself establish the exact limit.

COMMON TRAP

Conditional convergence is not absolute convergence

The exact value $\pi/2$ concerns the ordered improper integral $\lim_{R\to\infty}\int_0^R \sin x/x\,dx$. It does not assert that $\int_0^\infty |\sin x/x|dx$ is finite, and it does not permit arbitrary rearrangement of positive and negative area.

LESSON 07 / 07

Two rational tails can be evaluated without numerical approximation

For the first additional rational tail, partial fractions give

$$\frac{1}{(x^2+1)(x^2+4)} = \frac{1}{3}\left(\frac{1}{x^2+1} - \frac{1}{x^2+4}\right).$$

Therefore

$$\int_0^\infty \frac{dx}{(x^2+1)(x^2+4)} = \frac{1}{3}\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = \frac{\pi}{12}.$$

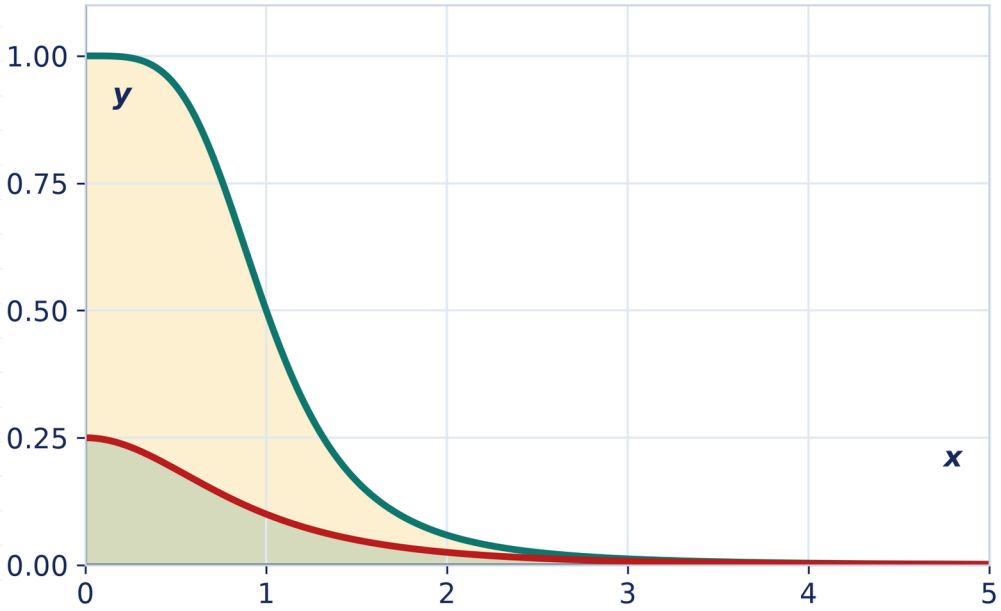
For the additional quartic tail, let $I = \int_0^\infty dx/(1+x^4)$. Inversion $x \mapsto 1/x$ gives $I = \int_0^\infty x^2 dx/(1+x^4)$. Hence

$$2I = \int_0^\infty \frac{1+x^2}{1+x^4}dx.$$

Divide by x^2 and put $u = x - 1/x$, so $du = (1+1/x^2)dx$ and u runs from $-\infty$ to ∞:

$$2I = \int_{-\infty}^{\infty} \frac{du}{u^2+2} = \frac{\pi}{\sqrt{2}}, \qquad I = \frac{\pi}{2\sqrt{2}}.$$

Two positive rational tails with exact areas



READ THE FIGURE
The teal curve 1/(1+x^4) has full tail area pi/(2 sqrt(2)); the red curve has full tail area pi/12. Only [0,5] is plotted, while the proofs include the remaining tails.

Log-sine uses reflection and a doubling substitution

For the additional log-sine integral, set

$$I=\int_0^{\pi/2}\ln\sin x\,dx.$$

The endpoint singularities are integrable. Reflection gives $I=\int_0^{\pi/2}\ln\cos x\,dx$, so

$$2I=\int_0^{\pi/2}\ln(\sin x\cos x)dx=\int_0^{\pi/2}\ln\sin(2x)dx-\tfrac{\pi}{2}\ln 2.$$

With $u=2x$ and symmetry on $[0,\pi]$,

$$\int_0^{\pi/2}\ln\sin(2x)dx=\tfrac{1}{2}\int_0^{\pi}\ln\sin u\,du=I.$$

Thus $2I=I-(\pi/2)\ln 2$, and

$$\int_0^{\pi/2}\ln\sin x\,dx=-\tfrac{\pi}{2}\ln 2.$$

A log-tangent consequence follows immediately:

$$\int_0^{\pi/2}\ln\tan x\,dx=\int_0^{\pi/2}\ln\sin x\,dx-\int_0^{\pi/2}\ln\cos x\,dx=0.$$

COMMON TRAP

Logarithmic identities require positive inputs inside each interval

Use $\ln(\sin x\cos x)$ on $(0,\pi/2)$, where both factors are positive, and interpret endpoints through limits. The identity $\ln(ab)=\ln a+\ln b$ is not applied at an endpoint where either logarithm is undefined.